

Supernatant Phase in Polyelectrolyte Complex Coacervation: Cluster Formation, Binodal, and Nucleation

Pengfei Zhang* and Zhen-Gang Wang*



Cite This: *Macromolecules* 2022, 55, 3910–3923



Read Online

ACCESS |



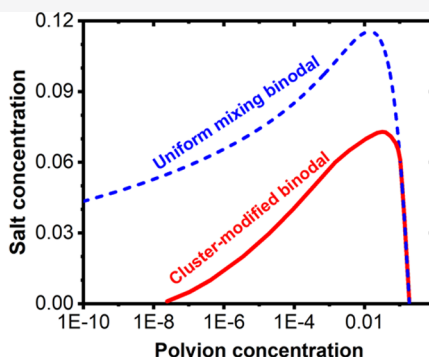
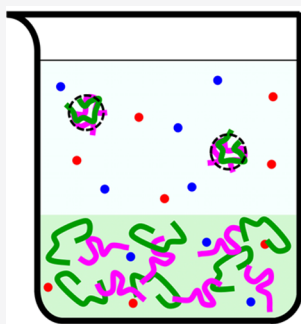
Metrics & More



Article Recommendations



Supporting Information



ABSTRACT: This work studies the structure and thermodynamics of the supernatant phase in polyelectrolyte complex coacervation, a relatively unexplored area. By combining the cluster theory in dilute solution with our recently developed mean-field theory for inhomogeneous polyelectrolyte solutions (Zhang, P.; Wang, Z.-G. *Macromolecules* 2021, 54, 10994), we systematically investigate the structure of finite-sized clusters formed by oppositely charged polyions in symmetric dilute solutions and how these clusters affect the binodal, spinodal, and nucleation for polyelectrolyte complex coacervation. We find that both the polyion concentration deep inside the cluster and the interfacial tension decrease with increasing the cluster size, reaching their respective bulk coexistence values with corrections inversely proportional to the cluster radius. The polyion concentration in the supernatant phase at coexistence is several orders of magnitude higher than that obtained under the uniform mixing approximation. For most relevant conditions away from the critical point, the supernatant phase consists predominantly of polyion pairs. By examining the nucleation barrier in supersaturated solutions, we can determine a pseudo-spinodal when the barrier is a few multiples of the thermal energy. The location of this pseudo-spinodal is similarly shifted to much higher concentrations than predicted under the uniform mixing approximation. Making the volume approximation for the clusters, we obtain simple analytical expressions for the cluster formation free energy, the modified binodal, and the pseudo-spinodal. In particular, we propose a simple approximate formula for estimating the concentration of the coexisting supernatant phase in terms of the chain length, interfacial tension, and the polyion concentration in the coacervate phase.

1. INTRODUCTION

Polyelectrolyte complex coacervates formed via an associative liquid–liquid phase separation by mixing oppositely charged polyelectrolyte (PE) solutions have a broad range of industrial applications, such as polymer film fabrication, hydrogel formation, and protein encapsulation.^{1–15} In protein therapeutics, for example, protein drugs can be encapsulated in the polypeptide complex coacervate droplets for effective and efficient drug delivery and subsequent release.^{5,8,13} Polyelectrolyte complex coacervates also have significant biological relevance due to their role in bioinspired adhesives, membraneless organelles, and artificial protocells.^{6,16–27} In recent years, considerable efforts have been devoted to understanding the structure and dynamics of polyelectrolyte complex coacervates, and significant progress has been made

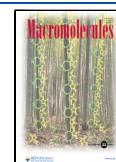
on both the theoretical and experimental fronts,^{28–84} we refer the readers to several recent excellent reviews.^{85–93}

In contrast to the coacervate phase, conspicuously less attention has been paid to the supernatant phase with which the PE-rich coacervate is in equilibrium. While the supernatant phase is too deficient of PEs to be of practical use, studying the structure and properties of this phase is still important for several reasons. First, varying the parameters in the supernatant phase provides a convenient and effective way to modulate the

Received: February 15, 2022

Revised: April 13, 2022

Published: May 12, 2022



physical properties of the coexisting coacervates. For example, increasing the salt concentration in the supernatant phase in general results in swelling of the coacervate and thus a decrease in the storage modulus.^{38,91,94} Second, quantitative and accurate determination of polymer concentration in the supernatant phase is an integral part of understanding the phase behavior of PE complex coacervation. For example, in cell biology, the saturation concentration that represents the polymer-poor branch of the binodal serves as a key physical quantity to characterize the minimum amount of proteins for forming a membraneless organelle.^{21,22} Finally, examination of the structure, energetics, and distribution of the clusters in the supernatant phase can provide valuable insights into the phase-transition kinetics in solutions of both synthetic and biological polyelectrolyte systems, i.e., whether phase separation proceeds via nucleation and growth or spinodal decomposition.^{95–97}

Due to the electrostatic attraction between oppositely charged polyions, polycations and polyanions in the supernatant phase are expected to form finite-sized polyion pairs or clusters, as first pointed out by Veis and co-workers in the 1960s.^{86,98} However, most of the theoretical investigations, such as the Voorn–Overbeek theory,^{2,37} random phase approximation,^{29,30,34,40} transition matrix theory,⁴⁴ and liquid-state theory,⁵⁴ assume that the polyions, small ions, and solvents are uniformly mixed, and do not account for these finite-sized PE clusters in the supernatant phase. These theories thus cannot properly describe the structure and thermodynamics of the supernatant phase. The situation is similar to uncharged homopolymer solutions, where the Flory–Huggins theory, which ignores the collapse of the polymers in poor solvents, predicts exponentially low monomer concentrations in the degree of polymerization for the polymer-poor branch of the binodal,⁹⁹ whereas a modified theory that accounts for the globule/clusters^{100–105} yields monomer concentrations that are many orders of magnitude higher than that predicted from the Flory–Huggins theory.^{104,106}

By combining the random phase approximation and the Lifshitz–Grosberg theory of polymer globules,^{100,107} Borue and Erukhimovich first examined the equilibrium structure of a symmetric neutral globule formed from one polycation chain and one polyanion chain, under the so-called volume approximation that is valid for a sufficiently large globule.¹⁰⁸ Subsequent work by Rumyantsev and Potemkin suggested that, in the absence of small ions, such neutral globules can form even when the polyion concentration is far below the supernatant binodal, i.e., when the solution is undersaturated with polyions.⁴² Based on the premise that all polycations and polyanions form neutral globules, Tainaka argued that polyelectrolyte complex coacervation results from the condensation of these globules due to electrostatic correlation.^{109–111} based on a virial expansion approach, while Adhikari et al. attributed the driving force for coacervation to the combined effect of dipolar and hydrophobic interactions.⁵⁰ Field-theoretical simulation by Delaney and Fredrickson clearly demonstrates that, for complex coacervation in the absence of small ions, oppositely charged polyions exist in the form of chain pairs and the polymer concentration in the supernatant phase is orders of magnitude higher than that predicted by the random phase approximation.⁴³ In systems with asymmetric polycation/polyanion chain length or concentration, the clusters can adopt more complex structures such as tadpoles¹¹² or cylinders/lamellae,¹¹³ in addition to

simpler spherical globules. Moreover, a series of computer simulation studies have been performed to examine the complexation behavior between oppositely charged polyions, and these studies show that the duplex formed by a pair of symmetric oppositely charged polyions roughly adopt a spherical structure.^{114–120}

In spite of these progresses, a systematic study on the cluster structure and energetics in the supernatant phase and how incorporation of the clusters affects the phase boundary and phase separation kinetics is lacking. In this work, we conduct such a study for the simplest symmetric coacervates where the clusters are expected to be electrically neutral and spherical. By generalizing our recently developed mean-field theory for inhomogeneous polyelectrolyte solutions,⁸² we first examine the structure, interfacial tension, and formation free energy of finite-sized clusters comprising equal numbers of polycations and polyanions. The formation free energy is then used to determine the cluster size distribution and the concentration of the coexisting supernatant phase. Our results show that away from the critical point, the clusters are predominantly in the form of single polyion pairs. The polyion concentration in the coexisting supernatant phase is shown to be several orders of magnitude higher than predicted from the uniform mixing approximation. We further use the cluster size distribution in the supersaturated state to calculate the free energy barrier for nucleation, from which we determine a pseudo-spinodal where the free energy barrier becomes order of the thermal energy. In Section II, we present the model and the general theoretical framework for calculating the cluster properties. The results are presented and discussed in Section III, where we first consider the salt-free system and then examine the effects of added salts. We conclude with a brief summary along with some outlook.

II. MODEL AND THEORETICAL FRAMEWORK

We consider a symmetric mixture of strongly charged polycations and polyanions in solution, with each polyion consisting of N coarse-grained monomers. For simplicity, we assume the counterions of the polyions to be the same as the salt ions, and all monomers, ions, and solvents to have the same volume v . Generalization to account for the size disparity among different species is straightforward. We further assume that all polyions are fully charged and both the monomers and salt ions are monovalent. Under appropriate conditions, the solution undergoes a liquid–liquid phase separation into a polymer-deficient supernatant phase and a liquid coacervate phase which contains most of the polymers. As alluded to in the Introduction, oppositely charged polyions are expected to form finite-sized clusters due to the electrostatic attraction—provided the polyion concentration is not too low and the salt concentration is not high—an effect that is not accounted for in most theoretical studies of the coacervation phase behavior. For symmetric mixtures, the clusters are expected to be electrically neutral, containing equal numbers of polycations and polyanions and equal numbers of small cations and anions. Unlike in ref 42, where the free energy was constructed based on a reference state of unpaired polycations and polyanions, we assume all chains are in the form of electrically neutral clusters and there are no unpaired polyions. We denote the cluster size by the number of polycations (or polyanions) m ; thus, an m -cluster contains m polycations and m polyanions. For the concentration range of interest in this work, the number density of these clusters, i.e., the number of m -clusters per unit volume, c_m , is sufficiently low and inter-cluster interactions can

be safely ignored. Thus, the system can be considered a dilute solution of these clusters. Let W_m be the free energy of an m -cluster, the total free energy of a system with the cluster number density c_m in a total volume $\mathcal{V}$ is

$$\mathcal{F} = \mathcal{V} \sum_{m=1}^{\infty} \{k_B T c_m [\ln(c_m v_m) - 1] + c_m W_m\} \quad (1)$$

where k_B is the Boltzmann constant and T is the temperature. The first term in the curly brackets represents the translational entropy of the m -clusters. The reference volume v_m arises from the evaluation of the partition function for internal degrees of freedom of the m -cluster;¹²¹ it requires accounting for the fluctuations in the cluster, a task that is beyond the mean-field theory presented in this work. Its precise value has only a minor effect on the thermodynamics of the system. For simplicity, we take it to be mNv ; with this choice, we have $c_m v_m = \phi_m$, which is simply the total volume fraction of the polycation (or polyanion) in the m -clusters. Minimizing $\mathcal{F}/\mathcal{V}$ with respect to c_m under the constraint of mass conservation $\sum_{m=0}^{\infty} m c_m = c_p$, where c_p is the total polycation (or polyanion) concentration in the system, we obtain

$$W_m + k_B T \ln \phi_m = 2N\mu_p m \quad (2)$$

where μ_p is the Lagrange multiplier from the mass conservation of polyions and the factor 2 arises from the fact that there are two mass conservation constraints with the same Lagrange multiplier due to the symmetry between the polycation and the polyanion. From this equation, we obtain the following cluster size distribution

$$\phi_m = \phi_1^m \exp[-\Delta W_m/k_B T] = \exp[-\Delta G_m/k_B T] \quad (3)$$

where ϕ_1 is the volume fraction of polycations (polyanions) in polyion pairs each containing one polycation and one polyanion (hereafter referred to simply as polyion pairs). $\Delta W_m \equiv W_m - mW_1$ represents the free energy difference between an m -cluster and m polyion pairs, excluding translational entropy of the clusters. In the second part of the equation, we define $\Delta G_m \equiv \Delta W_m - k_B T m \ln \phi_1$, which has the interpretation as the free energy change in coalescing m polyion pairs (i.e., $m = 1$) into an m -cluster, with the second term in ΔG_m accounting for the translational entropy loss of the polyion pairs.

The Lagrange multiplier μ_p in eq 2 is naturally interpreted as the chemical potential of the polyions in the cluster on a per-monomer basis. For undersaturated solutions, μ_p is determined by the total polyion concentration in the system. However, at the coexistence with the coacervate phase, μ_p must equal the chemical potential of the coacervate phase, which in turn determines the concentration of the polyions in the supernatant phase.

The free energy of the cluster, W_m is the key quantity in determining the cluster size distribution and other thermodynamic properties of the supernatant phase. Computer simulation^{115,117} of the complexation of a single polycation and a single polyanion showed that the two chains in a polyion pair form a well-mixed “scrambled eggs” structure¹²² as a spherical globule. We expect this to be true for any m . We thus consider a single m -cluster centered in a spherical volume of sufficiently large size containing the polyions with the small ions and solvent. Since the small ions and solvents in a cluster are in equilibrium with the outer reservoir, the volume is a semi-open system—open with respect to small ions and

solvent but closed with respect to the polyions. The free energy of the cluster W_m is the excess semi-grand potential, i.e., the semi-grand potential minus the grand potential of the reservoir without the polyions.

We apply the inhomogeneous mean-field theory developed in ref 82 to this spherical volume of a semi-open system containing an m -cluster. For a symmetric mixture with radial density profiles of polycations (or polyanions) and cations (or anions) given, respectively, by $\phi_p(r)$ and $\phi_s(r)$, connected to a reservoir with cation (or anion) chemical potential μ_{sr} corresponding to a cation (or anion) concentration ϕ_{sr} ¹²³ the semi-grand free energy functional reads

$$J_m = 4\pi \int_0^{R_\infty} r^2 dr \left[f_b(\phi_p(r), \phi_s(r)) + 2 \frac{k_B T b^2}{24v\phi_p(r)} \left(\frac{\partial \phi_p(r)}{\partial r} \right)^2 - 2 \frac{\mu_s}{v} \phi_s(r) \right] \quad (4)$$

where R_∞ is the radius of the spherical volume. Conceptually it can be taken as ∞ . For numerical calculations, we choose a value that is larger than the radius of cluster where the polyion density becomes vanishingly small and the small ion and solvent concentrations reach the reservoir condition; the exact value of R_∞ is immaterial for all of the properties we consider.

The first term in eq 4 represents the localized Helmholtz free energy density for a homogeneous solution and is given by⁸²

$$f_b = \frac{k_B T}{v} \left\{ \frac{2\phi_p}{N} \ln \phi_p + 2\phi_s \ln \phi_s + (1 - 2\phi_p - 2\phi_s) \ln(1 - 2\phi_p - 2\phi_s) + \frac{2l_b/\sigma}{1 + \kappa\sigma} \phi_p - \frac{1}{4\pi} \left[\ln(1 + \kappa\sigma) - \kappa\sigma + \frac{(\kappa\sigma)^2}{2} \right] \right\} \quad (5)$$

where $l_b = e^2/4\pi\epsilon_0\epsilon_r k_B T$ is the Bjerrum length with e , ϵ_0 , and ϵ_r being the elementary charge, the vacuum permittivity, and the dielectric constant of the solvent, respectively, and κ is the inverse of the Debye screening length given by $\kappa\sigma = \sqrt{8\pi(l_b/\sigma)(\phi_p + \phi_s)}$. The three terms in the first line account for the mixing entropy of polyions, small ions, and solvents, and the last term is the generalized Debye–Hückel correlation free energy¹²⁴ for disconnected ions with diameter $\sigma \equiv v^{1/3}$. We note that some authors excluded the polymer contribution in the mixing entropy^{100,108,125} on the ground that the center of mass is fixed for the cluster; here, we include this contribution so as to reproduce the bulk phase for a sufficiently large cluster. The difference between these two approaches is numerically inconsequential for large N . The fourth term is a first-order approximation for the chain connectivity contribution to the electrostatic correlation, where we have made a further simplification by dropping an $O(N^{-1})$ term; we refer interested readers to the detailed derivation in ref 82. Inclusion of this term allows capturing the correct negative tie-line slope in the polyion–salt phase diagram, which was missed in the Voorn–Overbeek theory or the generalized Voorn–Overbeek theory. The electrostatic correlations embodied by the last two terms in eq 5 provide the driving force

for phase separation. The second term in eq 4 is the well-known Lifshitz conformation entropy penalty due to inhomogeneous polymer density,^{107,126} where b is the Kuhn length of the polymer.

In this work, we focus on symmetric mixtures, for which the average local charge density is zero everywhere and there is no mean-field electrostatic contribution to J_m . For general asymmetric mixtures, there can be charge separation in the interfacial region and then J_m will include a mean-field electrostatic contribution; see ref 82 for details. Moreover, here we focus on the simplest model for polyelectrolyte complex coacervation, which includes only the electrostatic correlation, the chain connectivity, and excluded volume effects (accounted for by incompressibility); we ignore any χ interactions among polymers, solvents, and salt. Incorporation of these interactions will clearly lead to richer and more complex phenomena. However, a systematic investigation on the interplay between the electrostatic correlation and these χ interactions is beyond the scope of this work and deserves a separate study in future.

Minimizing J_m with respect to $\phi_p(r)$ and $\phi_s(r)$ under the constraint of the mass conservation of the polymer and with the salt reservoir chemical potential, we obtain the Euler–Lagrange equations for polyions and salts in the spherical coordinate as

$$\frac{\partial f_b(\{\phi_i(r)\})}{\partial \phi_p} - \frac{k_B T b^2}{6\nu \phi_p} \left[\frac{\partial^2 \phi_p(r)}{\partial r^2} + \frac{2}{r} \frac{\partial \phi_p(r)}{\partial r} \right] = \frac{\lambda_{p,m}}{\nu} \quad (6)$$

and

$$\frac{\partial f_b(\{\phi_i(r)\})}{\partial \phi_s} = \frac{\mu_s}{\nu} \quad (7)$$

respectively. Note that in eq 6 we have introduced the “wave function” $\varphi_p(r) \equiv \sqrt{\phi_p(r)}$, and the Lagrange multiplier $\lambda_{p,m}$ is used to enforce the mass conservation of polyions in the cluster, i.e.,

$$\frac{4\pi}{\nu} \int_0^{R_\infty} dr r^2 \varphi_p^2(r) = mN \quad (8)$$

$\lambda_{p,m}$ has the natural interpretation as the chemical potential of a polycation (polyanion) in the m -cluster on a monomer basis. At a specified salt reservoir concentration $\phi_{s,r}$ and for a given m , the concentration profiles $\phi_p(r)$ and $\phi_s(r)$ in the cluster and $\lambda_{p,m}$ are calculated by numerically solving eqs 6–8 using the finite difference method. Substituting the $\phi_p(r)$ and $\phi_s(r)$ so obtained into eq 4, we obtain the semi-grand free energy for the spherical volume containing the m -cluster. This semi-grand free energy obviously includes contributions from the salt reservoir and depends on the size of the spherical volume R_∞ . To extract the free energy associated with the m -cluster, we note that in the absence of the polyions, the grand potential of the salt reservoir with the same volume is

$$\begin{aligned} J_{s,r} &= 4\pi \int_0^{R_\infty} r^2 dr \left[f_b(\phi_p = 0, \phi_{s,r}) - 2\frac{\mu_s}{\nu} \phi_{s,r} \right] \\ &= -\frac{4\pi R_\infty^3}{3} \Pi_r \end{aligned} \quad (9)$$

where $\Pi_r = -[f_b(\phi_p = 0, \phi_{s,r}) - 2\mu_s \phi_{s,r}/\nu]$ is the osmotic pressure of the salt reservoir. The salt concentration in the

reservoir $\phi_{s,r}$ is related to the chemical potential μ_s through eq 7 by setting $\phi_p = 0$. The free energy of the m -cluster is then

$$\begin{aligned} W(m, \mu_s) &= J_m - J_{s,r} \\ &= 4\pi \int_0^{R_\infty} dr r^2 \left[f_b(\{\phi_i(r)\}) + \frac{k_B T b^2}{3\nu} \left(\frac{\partial \varphi_p(r)}{\partial r} \right)^2 \right. \\ &\quad \left. - 2\frac{\mu_s}{\nu} \phi_s(r) + \Pi_r \right] \end{aligned} \quad (10)$$

For sufficiently large N , the cluster will consist of a nearly uniform core and an interfacial region. To calculate the interfacial tension for an m -cluster, we identify the surface of tension^{127,128} by the following method. We define a reference system consisting of a uniform core separated from the salt reservoir by a sharp dividing surface at R (the surface of tension, yet to be determined). Inside the core, the polyion density $\phi_{p,c}$ and the salt-ion density $\phi_{s,c}$ are determined by solving $\partial f_b/\partial \phi_{p,b} = \lambda_{p,m}/\nu$ and $\partial f_b/\partial \phi_{p,b} = \mu_s/\nu$, respectively, where $\lambda_{p,m}$ is the Lagrange multiplier obtained earlier. For a given choice of R , the interfacial tension is the excess grand free energy with respect to the reference system

$$\gamma(R) = \frac{1}{4\pi R^2} \left[\Omega(m, \mu_s) - \frac{4\pi R^3}{3} \omega_c - \frac{4\pi(R_\infty^3 - R^3)}{3} \omega_r \right] \quad (11)$$

where $\Omega(m, \mu_s)$ is the grand potential of the actual system consisting of the cluster and the reservoir, given by

$$\begin{aligned} \Omega(m, \mu_s) &= 4\pi \int_0^{R_\infty} dr r^2 \left[f_b(\phi_p(r), \phi_s(r)) \right. \\ &\quad \left. + \frac{k_B T b^2}{3\nu} \left(\frac{\partial \varphi_p(r)}{\partial r} \right)^2 - 2\frac{\mu_s}{\nu} \phi_s(r) - 2\frac{\lambda_{p,m}}{\nu} \phi_p(r) \right] \end{aligned} \quad (12)$$

and $\omega_c = f_b(\phi_{p,c}, \phi_{s,c}) - 2\lambda_{p,m}\phi_{p,c}/\nu - 2\mu_s\phi_{s,c}/\nu$ and $\omega_r = f_b(\phi_{p,r}, \phi_{s,r}) - 2\mu_s\phi_{s,r}/\nu$ in eq 11 are, respectively, the grand potential density of the uniform core of the reference system and the grand potential density of the reservoir. From thermodynamics, the osmotic pressure of the inner core $\Pi_c = -\omega_c$ and that of the reservoir $\Pi_r = -\omega_r$. The location of the surface of tension R^* is then determined by the condition $\partial \gamma(R)/\partial R|_{R=R^*} = 0$, which yields

$$R^* = \left[\frac{3}{2\pi} \frac{\Omega(m, \mu_s) - V\omega_r}{\Pi_c - \Pi_r} \right]^{1/3} \quad (13)$$

with $V = 4\pi R_\infty^3/3$ being the volume of the system containing the m -cluster and the reservoir. Substituting R^* into eq 11, we find that the surface tension $\gamma_m \equiv \gamma(R^*)$ satisfies the Laplace equation

$$\frac{2\gamma_m}{R^*} = \Pi_c - \Pi_r \quad (14)$$

Note that γ_m depends on the cluster size and can be different from that for the planar interface. Moreover, from eqs 12 and 10, we have $\Omega(m, \mu_s) - V\omega_r = W(m, \mu_s) - 2mN\lambda_{p,m}$, and eq 13 can be equivalently written as

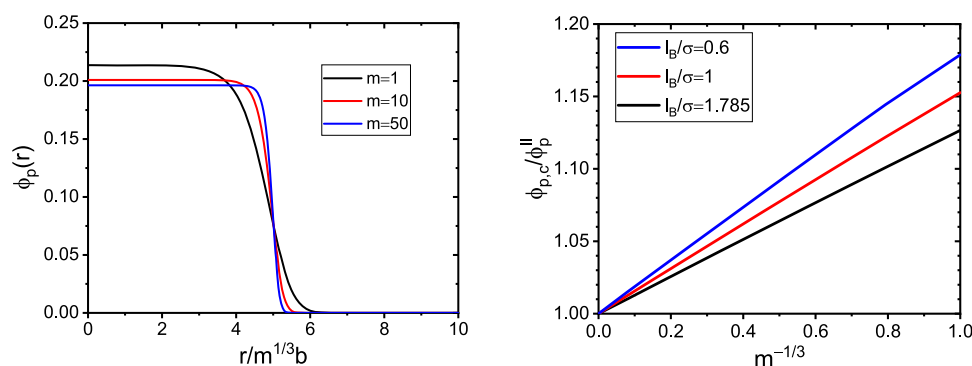


Figure 1. (a) Polycation (or polyanion) concentration profile $\phi_p(r)$ along the normalized radial distance $r/m^{1/3}b$ for finite-sized clusters consisting of $m = 1, 10$, and 50 pairs of polyions. (b) Polycation (polyanion) concentration in the uniform core of the cluster $\phi_{p,c}$ as a function of $m^{-1/3}$ for three l_B/σ . $\phi_p^{\text{II}} \approx 0.214, 0.201$, and 0.196 for $l_B/\sigma = 1.785, 1$, and 0.6 , respectively.

$$R^* = \left[\frac{3}{2\pi} \frac{W(m, \mu_s) - 2mN\lambda_{p,m}}{\Pi_c - \Pi_r} \right]^{1/3} \quad (15)$$

For very large clusters ($m \rightarrow \infty$), the thermodynamics is dominated by the uniform core contribution and the surface effects become negligible. Under the so-called volume approximation first proposed by Lifshitz in the context of coil–globule transition for neural polymers,^{100,106,126} the equilibrium concentration of polyions and salt in the cluster can be simply determined by the equality of the salt-ion chemical potential μ_s and the equality of osmotic pressure Π , respectively, between the reservoir and the core of the cluster, by the simultaneous equations, eq 7 and

$$\begin{aligned} f_b(\phi_{p,c}, \phi_{s,c}) - 2\frac{\lambda_{p,\infty}}{\nu}\phi_{p,c} - 2\frac{\mu_s}{\nu}\phi_{s,c} \\ = f_b(\phi_{p,c} = 0, \phi_{s,r}) - 2\frac{\mu_s}{\nu}\phi_{s,r} \end{aligned} \quad (16)$$

where the Lagrange multiplier $\lambda_{p,\infty}$, upon ignoring the spatial derivatives in eq 6, is now given by

$$\frac{\lambda_{p,\infty}}{\nu} = \frac{\partial f_b(\{\phi_i(r)\})}{\partial \phi_p} \quad (17)$$

It is natural to regard an infinitely large cluster as the bulk coacervate phase. Thus, the polyion concentration $\phi_{p,c}$ and salt-ion concentration $\phi_{s,c}$ in the core of an infinitely large cluster can be considered the equilibrium coexisting concentrations ϕ_p^{II} and ϕ_s^{II} of the coacervate phase.¹²⁹ These phase equilibrium conditions under the volume approximation for symmetric complex coacervation were first proposed by Borue and Erukhimovich¹⁰⁸ and later used by Romyantsev, Zhulina, and Borisov.⁵¹ (As mentioned earlier, there is a slight difference between our current theoretical formulation and that in refs 108 and 51 regarding the polyion contribution to the mixing entropy: our f_b in eqs 16 and 17 includes this contribution, while refs 108 and 51 excluded it. Both approaches can be justified: ignoring this term is reasonable from the point of view of a polyion pair or small clusters, while keeping it allows the large cluster to be treated consistently as a bulk phase. Since the contribution from this term to the osmotic pressure is rather small for large N , we do not consider this difference important.) For the salt-free case, $\phi_{s,r} = \phi_s^{\text{II}} = 0$, and thus, ϕ_p^{II} can be determined simply by solving $\Pi(\phi_p^{\text{II}}) = 0$.

From eq 17, $\lambda_{p,\infty}$ is the chemical potential of the polyions in the coacervate phase μ_p^{II} . By equating this chemical potential with the chemical potential in eq 2, we determine the cluster size distribution ϕ_m at the coexistence. The polyion concentration on the supernatant binodal is then $\phi_p^{\text{I}} = \sum_{m=1} \phi_m$.

III. RESULTS AND DISCUSSION

In our model system, complex coacervation is driven by the electrostatic correlation whose strength is characterized by l_B/σ . For an aqueous solution at room temperature, $l_B \approx 0.714$ nm. When taking into account the temperature dependence of the dielectric constant of a pure solvent ϵ_r , l_B increases with temperature and thus our theory can capture the lower critical solution temperature behavior of complex coacervation.^{61,83} However, for the purpose of this work, we do not pursue this explicit temperature dependence; instead, we regulate the electrostatic correlation strength by varying l_B/σ , which can be achieved experimentally by varying the temperature, solvent type, or the segment size. In this section, we first examine the symmetric solution without small ions. After calculating the structure and formation free energy for clusters consisting of different pairs of polyions, we construct the cluster-corrected binodal and spinodal curves in the ϕ_p – l_B/σ plane. To understand the numerical results, we perform simplified analysis of the formation free energy, supernatant binodal, and nucleation, based on the volume approximation for the polyion pair. We then examine how the addition of small ions affects the cluster structure and the corresponding formation free energy, and how the incorporation of clusters affects the binodal and spinodal in the polymer–salt concentration (i.e., ϕ_p – ϕ_s) phase diagram. In the numerical calculations, we set $N = 100$ and $b = \sigma$.

III.I. Salt-Free Case. III.I.I. Structure and Interfacial Tension of Finite-Sized Clusters. We start with examining the structure and interfacial tension of the finite-sized clusters formed in a dilute solution of symmetric mixtures of polycations and polyanions in the absence of small ions. In Figure 1a, we show the radial polycation (polyanion) concentration profile $\phi_p(r)$ for clusters consisting of $m = 1, 10$, and 50 polyion pairs, respectively, for $l_B/\sigma = 1.785$ (corresponding to $\sigma = 0.4$ nm). In all cases, $\phi_p(r)$ consists approximately of an inner uniform core and an outer diffusive surface layer where $\phi_p(r)$ decreases rapidly toward zero. Note that the polycation concentration in the uniform core is simply $\phi_{p,c}$ as defined by the reference state when calculating the

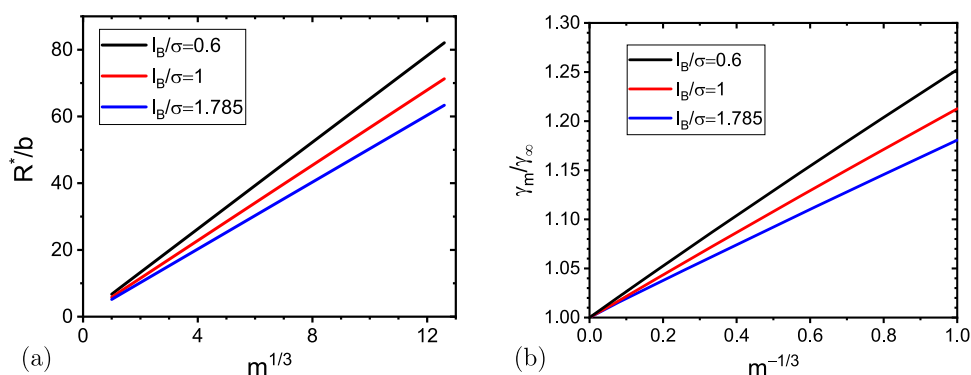


Figure 2. (a) Location of the surface of tension R^* as a function of $m^{1/3}$, and (b) the interfacial tension γ_m as a function of $m^{-1/3}$ for clusters at three l_B/σ . $\gamma_\infty = \lim_{m \rightarrow \infty} \gamma_m \approx 0.051$, 0.026 , and 0.012 for $l_B/\sigma = 1.785$, 1 , and 0.6 , respectively.

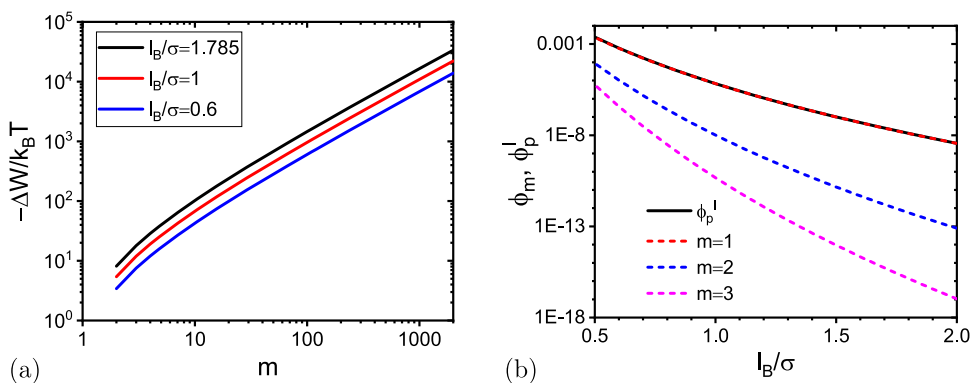


Figure 3. (a) Free energy change $\Delta W_m = W_m - mW_1$ of an m -cluster relative to m polyion pairs for three l_B/σ . (b) Polycation concentration in a single polyion pair (i.e., ϕ_1), in clusters with $m = 2$ and 3 (denoted by ϕ_2 and ϕ_3 , respectively), and the total polycation concentration $\phi_p^I = \sum_m \phi_m$ on the supernatant binodal.

interfacial tension. With increasing m , $\phi_{p,c}$ decreases gradually and approaches ϕ_p^{II} in the limit of $m \rightarrow \infty$. Moreover, the surface layer for the different m -clusters is located approximately at the same $r/m^{1/3}b$. Finally, we remark that the cluster structure obtained in our work—with a nearly uniform core separated by an interfacial region—is markedly different from the Gaussian density distribution assumed by Tainaka.^{109–111}

As a quantitative characterization for the m -dependence of $\phi_{p,c}$, Figure 1b shows that $\phi_{p,c} = \phi_p^{\text{II}}(1 + C_p m^{-1/3})$ for large m , where $C_p > 0$ is a constant that increases with decreasing l_B/σ . This m -dependence is a reflection of the Laplace pressure effect (cf. eq 15) since the osmotic pressure inside a finite-sized droplet is higher than that of the infinitely large cluster (which in turn equals that of the surrounding reservoir) and the difference is inversely proportional to the radius of the droplet.^{127,130} To maintain a higher osmotic pressure for a smaller cluster, a larger polymer concentration $\phi_{p,c}$ is required.

The behavior of $\phi_p(r)$ for clusters at other values of l_B/σ is qualitatively similar. Quantitatively, with decreasing l_B/σ , the core density $\phi_{p,c}$ decreases and the interfacial layer becomes more diffusive; see Figure S1a for details. These phenomena are expected due to the weakening of electrostatic correlation with decreasing l_B/σ .

Figure 2a,b shows, respectively, the locations of the surface of tension R^* and the interfacial tension γ_m for the clusters. At a given l_B/σ , R^* is nearly proportional to $m^{1/3}$, i.e., $R^* \approx C_R m^{1/3}$, where the coefficient C_R is larger for smaller l_B/σ , consistent with the decreased density at decreased electrostatic correlation. On the other hand, γ_m is larger for smaller m and

$\gamma_m \approx \gamma_\infty (1 + m^{-1/3} C_\gamma)$, where $\gamma_\infty \equiv \lim_{m \rightarrow \infty} \gamma_m$ and $C_\gamma > 0$ is a constant. By invoking the relation $R^* \propto m^{1/3}$, we then obtain $\gamma_m \approx \gamma_\infty (1 - 2l_T/R^*)$, where l_T is the Tolman length^{128,131} that characterizes the curvature correction. From Figure 1b, we see that $l_T < 0$ and its magnitude $|l_T|$ increases with decreasing l_B/σ .

III.III. Formation Free Energy of Clusters, Binodal, and Spinodal. The free energy of formation $\Delta W_m \equiv W_m - mW_1$ in eq 3 is the free energy change between an m -cluster and m polyion pairs (without the translational entropy of the clusters) and is the driving force for coalescing m polyion pairs into a single m -cluster. When ΔW_m is sufficiently negative to overcome the contribution due to the translational entropy of polyion pairs, large clusters will form spontaneously; otherwise, polyions will exist in the form of small polyion pairs. In Figure 3a, we present ΔW_m for three representative l_B/σ . At each l_B/σ , ΔW is approximately proportional to m at large m . Moreover, at a given m , $|\Delta W_m|$ is larger for higher l_B/σ because of the stronger electrostatic correlation and hence the larger interfacial tension.

As explained in Section II, at a given l_B/σ , the polyion concentration in the coacervate phase ϕ_p^{II} is determined by $\Pi(\phi_p^{\text{II}}) = 0$. From ϕ_p^{II} , we calculate μ_p in this phase by eq S1. Along with W_m , we can calculate the concentration of polyions ϕ_m for the different clusters and then determine the polycation concentration in the supernatant binodal as $\phi_p^I = \sum_m \phi_m$. The l_B/σ dependence of μ_p and W_1 is given in Figure S1b. In Figure 3b, we show ϕ_m for clusters with $m = 1, 2$, and 3 , and the total polyion concentration in the supernatant phase ϕ_p^I , as functions

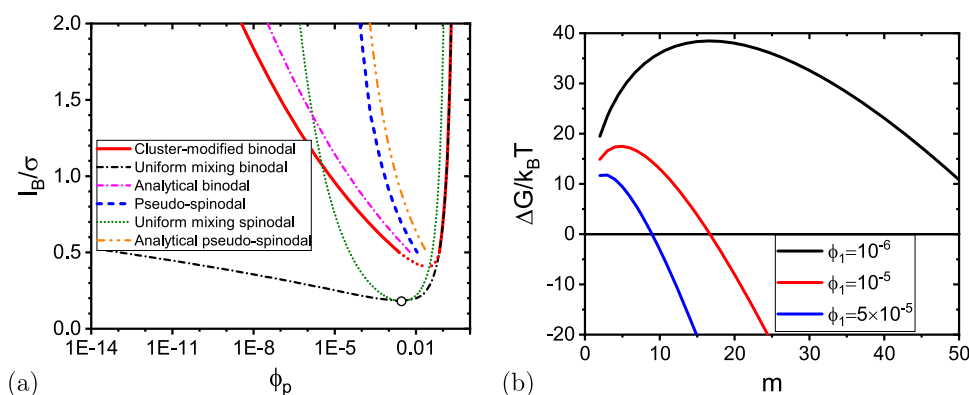


Figure 4. (a) Phase diagram in the ϕ_p – l_B/σ plane. The red curve is the numerical result for the binodal; the magenta curve is the result from using the approximate expression, eq 19. The blue curve is the numerical result for the pseudo-spinodal determined by setting $\Delta G^* = 10k_B T$; the orange curve is the result from using the approximate expression, eq 21. The black and olive curves are, respectively, the binodal and spinodal under the uniform mixing approximation. (b) The formation free energy of an m -cluster, $\Delta G = \Delta W - k_B T m \ln \phi_1$, as a function of m for three representative supersaturated cases ϕ_1 ; $l_B/\sigma = 1.785$ and $\phi_p^I \approx 1.39 \times 10^{-8}$.

of l_B/σ . For the range of l_B/σ values examined, the overall polyion concentration in the supernatant phase is dominated by the single polyion pairs (i.e., $m = 1$), as seen by the overlap between the black solid curve and the red dashed curve. For all m , ϕ_m decreases with increasing l_B/σ , indicating that the polyion concentration in clusters in the supernatant phase at coexistence becomes more dilute for larger l_B/σ .

By collecting ϕ_p^{II} and ϕ_p^{I} at different values of l_B/σ , we construct the binodal in the ϕ_p – l_B/σ plane, as shown by the red solid curve in Figure 4a. At sufficiently low l_B/σ , the ion concentration profile in the clusters becomes rather diffusive and different clusters may start to interact with each other as their numbers become more numerous. In addition, long-range critical fluctuation becomes significant near the critical point, the description of which requires advanced techniques such as renormalization group theory.¹³² Such a pursuit is beyond the scope of this work. Nevertheless, for completing the phase diagram, in Figure 4a, we qualitatively sketch the missing part of the binodal close to the critical point by the red dotted curve, by visual continuation of the two branches of the red solid curve. For comparison, in Figure 4a, we include the binodal obtained under the uniform mixing approximation (the black dashed-dotted curve), calculated by solving the phase equilibrium condition (i.e., the equality of the polymer chemical potential and osmotic pressure in the two coexisting phases, respectively) from the bulk free energy density given by eq 5. We see that the coacervate branch (i.e., ϕ_p^{II}) from both methods is nearly the same. However, the supernatant branch (i.e., ϕ_p^{I}) accounting for the clusters is many orders of magnitude larger than that from the uniform mixing approximation, and the difference is more drastic at larger l_B/σ . These features are qualitatively consistent with the field-theoretical simulation study by Delaney and Fredrickson.⁴³ Moreover, while we are unable to accurately locate the critical point in the cluster model, from the general trend of the supernatant binodal, we expect the critical point to shift to larger l_B/σ relative to the prediction from the uniform mixing approximation (the black open circle in Figure 4a).

At a given l_B/σ , when the polycation concentration ϕ_p exceeds the supernatant binodal ϕ_p^{I} , a homogeneous solution becomes supersaturated and phase separation is expected to occur via the nucleation and growth mechanism. Assuming that there is no significant barrier in the fusion of clusters, we

can calculate the nucleation barrier by examining the free energy landscape using the cluster size (i.e., m) as the reaction coordinate. The size of the cluster at the maximum, denoted by m^* hereafter, corresponds to the critical nucleus size. Taking $l_B/\sigma = 1.785$ as an example, in Figure 4b, we then show our numerical results on the formation free energy of the cluster ΔG as a function of m at several representative values of supersaturation $\phi_1 > \phi_p^{\text{I}} \approx 1.39 \times 10^{-8}$. We see that, with increasing ϕ_1 , both the formation free energy barrier ΔG^* and the critical nucleus size m^* decreases. When the free energy barrier becomes comparable to the thermal energy $k_B T$, the metastable homogeneous state becomes no longer kinetically viable and phase separation will proceed spontaneously via spinodal decomposition. While a rigorous criterion to distinguish the nucleation and growth regime and the spinodal decomposition regime is lacking, we choose an operational criterion $\Delta G^* = 10k_B T$, following previous practices for homopolymer solutions¹⁰⁴ and polymer blends,^{133–136} to define the *effective* metastability limit or pseudo-spinodal. Hereafter, we term the special value of ϕ_1 at which $\Delta G^* = 10k_B T$ as the pseudo-spinodal concentration ϕ_p^{S} .

The blue dashed curve in Figure 4a shows the pseudo-spinodal curve so-constructed in the ϕ_p – l_B/σ plane. For comparison, we also include the spinodal under the uniform mixing approximation (the green dotted curve), calculated by the condition $\partial^2 f_b(\phi_p)/\partial \phi_p^2 = 0$. We see that, at a given l_B/σ , the pseudo-spinodal calculated from the cluster model is also several orders of magnitude larger than that from the uniform mixing approximation; however, the difference is less drastic than for the binodal. We note that for uncharged homopolymer solutions, the pseudo-spinodal concentration on the dilute side is actually *lower* than the mean-field spinodal predicted by the uniform (random) mixing (i.e., Flory–Huggins) approximation,¹⁰⁶ which indicates a fundamental difference between these two systems.

III.I.III. Approximate Analysis of the Formation Free Energy, Binodal, and Nucleation. To help understand our numerical results and to provide simple analytical expressions, here we perform an analysis of ΔW as well as its implications on the supernatant binodal and nucleation properties, based on the volume approximation of the properties of the clusters. First, using our definition of the surface of tension, we can write the free energy as $W_m = w_c(\phi_{p,c})V_m + 2mN\mu_p(\phi_{p,c}) +$

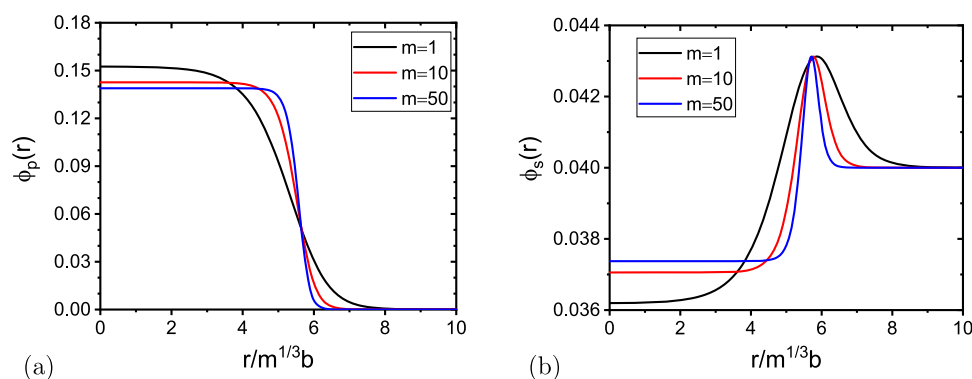


Figure 5. (a) Polycation (or polyanion) concentration profile $\phi_p(r)$ and (b) cation (anion) concentration profile $\phi_s(r)$ along the scaled radial distance $r/m^{1/3}$ for finite-sized clusters in equilibrium with a salty reservoir with $\phi_{sr} = 0.04$ and $l_B/\sigma = 1.785$.

$\gamma_m A_m$ with $V_m = 4\pi R^{*3}/3$ and $A_m = 4\pi R^{*2}$ being the volume and surface area of the reference system of the m -cluster, respectively. Under the volume approximation, the density of the core is approximately that of the bulk coacervate phase, i.e., $\phi_{p,c} = mN/V_m \approx \phi_p^{\text{II}}$, and we ignore the curvature correction in the interfacial tension by setting $\gamma_m \approx \gamma_\infty$. The grand potential density at polyanion density ϕ_p^{II} is $w_c(\phi_p^{\text{II}}) = f_b(\phi_p^{\text{II}}) - 2\phi_p^{\text{II}}\mu_p(\phi_p^{\text{II}})$. By invoking $(4\pi/3)R^{*3}\phi_p^{\text{II}} \approx mNv$, we then have $R^* \approx C_R m^{1/3}$ with $C_R \equiv (3Nv/4\pi\phi_p^{\text{II}})^{1/3}$, and $W_m = (4\pi/3)C_R^3 f_b(\phi_p^{\text{II}})m + 4\pi C_R^2 \gamma_\infty m^{2/3}$; thus

$$\Delta W_m = W_m - mW_1 = 4\pi C_R^2 \gamma_\infty (m^{2/3} - m) \quad (18)$$

Clearly, $\Delta W_m < 0$ for $m > 1$ and $\Delta W_m \propto m$ at large m , qualitatively consistent with our numerical results in Figure 3a. Further, the form of ΔW_m suggests that the reduction of the interfacial area is the main driving force for the coalescence of small clusters into larger clusters and eventually into a macroscopic coacervate phase.

From ΔW_m and eq 3, we further obtain $\Delta G_m \approx (-k_B T \ln \phi_1 - 4\pi C_R^2 \gamma_\infty)m + 4\pi C_R^2 \gamma_\infty m^{2/3}$. When the coefficient of the linear term is positive, formation of clusters with size $m > 1$ is unfavorable, and the system is undersaturated with the polyions existing in pairs (i.e., $m = 1$); when the coefficient is negative, the system favors a macroscopically large cluster (i.e., $m \rightarrow \infty$). Thus, the vanishing of this coefficient in this analysis provides an estimate of the concentration of polyion pairs at the coexistence. At this condition, $\Delta G_m \approx 4\pi C_R^2 \gamma_\infty m^{2/3} = -k_B T m^{2/3} \ln \phi_1$. Substituting it into eq 3, we obtain $\phi_m \approx \phi_1^{m^{2/3}}$. This simple relation suggests that the concentration of polyions in clusters with large m , ϕ_m , is much smaller than ϕ_1 , and thus, the supernatant phase is dominated by simple polyion pairs (i.e., $m = 1$), as shown by our numerical data in Figure 3b. Therefore, ϕ_1 at this condition is essentially the polyion concentration in the supernatant phase at coexistence ϕ_p^1 and we thus have

$$\begin{aligned} \phi_p^1 &\approx \exp(-4\pi C_R^2 \gamma_\infty / k_B T) \\ &= \exp[-(36\pi)^{1/3} (Nv/\phi_p^{\text{II}})^{2/3} \gamma_\infty / k_B T] \end{aligned} \quad (19)$$

This relation suggests a convenient estimate of the supernatant binodal ϕ_p^1 via ϕ_p^{II} and γ_∞ , the latter being more easily and accurately measurable in experiments. The magenta dashed-dotted curve in Figure 4a shows that this analytical binodal can indeed reasonably describe the numerical supernatant binodal (i.e., the red solid curve in Figure 4a). The discrepancy

between the two curves is most likely due to neglecting the curvature corrections in $\phi_{p,c}$ and γ_m when deriving the analytical expression of ΔW_m . In addition, a more complete analysis using the equality of the chemical potential will introduce a prefactor involving the concentration of the coacervate. Thus, eq 19 is more suitably viewed as an order-of-magnitude estimate rather than a quantitative one.

The formation free energy ΔG_m also provides valuable insights into the nucleation and growth mechanism when the solution is supersaturated, i.e., when $\phi_1/\phi_p^1 > 1$. After extremization of ΔG with respect to m , we obtain $m^* = 2S_0/(\ln S)^3$, where $S \equiv \phi_1/\phi_p^1$ is the supersaturation, and the parameter $S_0 \equiv (16\pi/3)N^2(\gamma_\infty v^{2/3}/k_B T)^3(\phi_p^{\text{II}})^{-2}$. The free energy barrier is then

$$\Delta G^* \equiv \Delta G(m^*) = k_B T S_0 / (\ln S)^2 \quad (20)$$

This is the nucleation barrier from the classical nucleation theory, which is valid when the supersaturation is not too high; the validity of the classical nucleation theory has recently been reported for multicomponent biomolecular condensates.⁹⁷

Note that S_0 is proportional to the square of chain length and the cube of surface tension and inversely proportional to the square of ϕ_p^{II} . These dependences are the same as for uncharged homopolymer solutions.¹⁰⁴ Furthermore, setting $\Delta G^* = 10k_B T$, we arrive at an analytical approximation of the pseudo-spinodal, i.e.,

$$\phi_p^s \approx \phi_p^1 \exp[1.294 Nv(\gamma_\infty/k_B T)^{3/2} (\phi_p^{\text{II}})^{-1}] \quad (21)$$

Figure 4a shows that this analytical expression provides a reasonable approximation (orange dashed-dotted curve) for the numerical pseudo-spinodal (the blue dashed curve).

We remark that most of the analysis given in this subsection for the salt-free system carries through to the system with salt. Furthermore, the main results are derived without resorting to specific coacervation theories, so they should be quite general.

III.II. Effects of Salt Concentration. Our discussion on the effects of salt ions parallels the discussion on the salt-free system, but here we will only present the numerical results. We first examine the structure and interfacial tension of the m -clusters and then study how incorporation of the clusters affects the binodal and spinodal in the polymer–salt concentration (i.e., ϕ_p – ϕ_s) plane. Since the effects of l_B/σ have already been discussed for the case without small ions and the qualitative effects remain similar, in this section, we take a fixed $l_B/\sigma = 1.785$ to focus on the salt concentration effects.

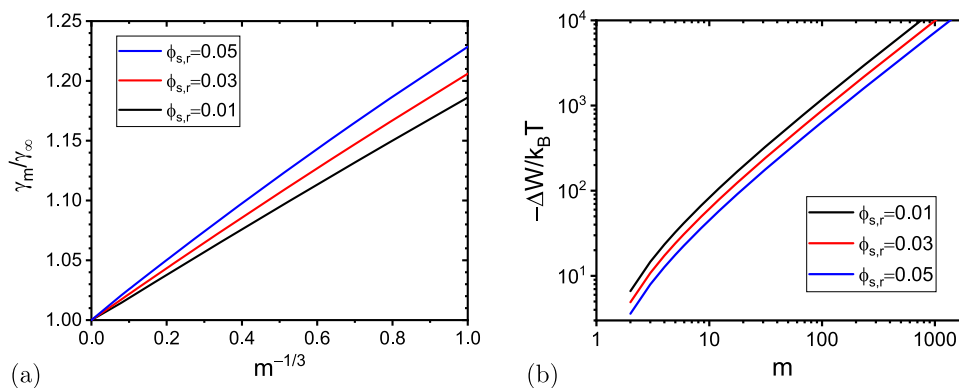


Figure 6. (a) Interfacial tension γ_m of finite-sized clusters as a function of $m^{-1/3}$ for three $\phi_{s,r}$; here, $\gamma_\infty \equiv \lim_{m \rightarrow \infty} \gamma_m \approx 0.038, 0.025$, and 0.016 for $\phi_{s,r} = 0.01, 0.03$, and 0.05 , respectively. (b) Formation free energy $\Delta W_m = W_m - mW_1$ of an m -cluster for three $\phi_{s,r}$, $l_B/\sigma = 1.785$ and $N = 100$.

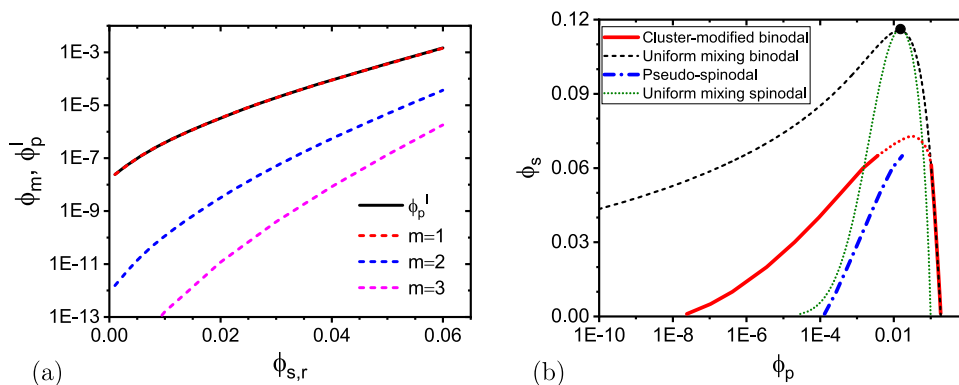


Figure 7. (a) Polycation concentration in the m -cluster with $m = 1, 2$, and 3 , ϕ_m , and the total polycation concentration ϕ_p^I on the supernatant binodal. (b) Phase diagram in the ϕ_p - ϕ_s plane at $N = 100$ and $l_B/\sigma = 1.785$. The red and black curves are, respectively, the cluster-modified binodal and the binodal under the uniform mixing approximation. The blue and olive curves are, respectively, the pseudo-spinodal and the spinodal under uniform mixing approximation.

III.III. Structure and Interfacial Tension of Finite-Sized Clusters. Figure 5a shows the concentration profile of polycation (polyanion) $\phi_p(r)$ for clusters with different m when the salt concentration in the reservoir $\phi_{s,r} = 0.04$. Similar to the case without small ions (i.e., Figure 1a), $\phi_p(r)$ for all m also has a uniform core with $\phi_p(r) = \phi_{p,c}$ and an outer diffusive surface layer. Likewise, $\phi_{p,c}$ is higher for smaller m due to the Laplace pressure effect. Quantitative comparison between Figures 5a and 1a suggests that the addition of salt leads to a decrease in $\phi_{p,c}$ and a thickening of the surface layer.

In Figure 5b, we show the cation (anion) concentration $\phi_s(r)$ as a function of the scaled radial distance $r/m^{1/3}b$ for clusters with different m . Note that $\phi_s(r)$ in the uniform core of the cluster is the same as the concentration $\phi_{s,c}$ defined by the reference state. We see that $\phi_{s,c}$ in the cluster core is in general less than the salt concentration in the reservoir $\phi_{s,r}$, qualitatively consistent with the bulk salt partitioning behavior reported in our previous work.^{53,54,82} Since the structure inside the uniform core of the cluster resembles that of the bulk coacervate phase, the reduced salt concentration in the cluster, like in the bulk coacervate phase, can be attributed to the chain connectivity contribution in the electrostatic free energy, following the analysis given in ref 53. With increasing m , $\phi_{s,c}$ increases slightly and approaches the coacervate concentration at coexistence ϕ_s^{II} in the limit of $m \rightarrow \infty$. Notably, we observe an enrichment of salt in the surface layer of polycation cluster, but the peak value is roughly independent of m . Moreover, the salt enrichment in the interfacial layer becomes weaker with

increasing $\phi_{s,r}$ (see Figure S2b). This behavior is qualitatively consistent with that for the planar interface as reported in our previous work,⁸² and presumably arises from the chain conformational entropy at the interface. Similar to the salt-free case, quantitative study on the m -dependence of $\phi_{p,c}$ and $\phi_{s,c}$ as shown by Figure S3, reveals that $\phi_{p,c} = \phi_p^{\text{II}}(1 + D_p m^{-1/3})$ and $\phi_{s,c} = \phi_s^{\text{II}}(1 - D_s m^{-1/3})$ at large m , where $D_p > 0$ and $D_s > 0$ are constants that increase with $\phi_{s,r}$.

From the structure of an m -cluster, we then calculate the free energy W_m , from which we identify the surface of the interface R^* and the interfacial tension γ_m following eqs 10, 14, and 15. The m -dependence of these quantities is qualitatively similar to that for the salt-free cases; here, we only present γ_m for demonstration. In Figure 6a, we show γ_m as a function of $m^{-1/3}$ for three values of the reservoir concentration $\phi_{s,r}$. Again, we see that $\gamma_m = \gamma_\infty(1 + D_\gamma m^{-1/3})$ for large m where the slope $D_\gamma > 0$ increases with the salt reservoir concentration $\phi_{s,r}$. This behavior also indicates that the interfacial tension of finite-sized cluster increases on decreasing the cluster size, and the leading correction is inversely proportional to the radius.

III.III.III. Formation Free Energy of Clusters, Binodal, and Spinodal. Next, we analyze the m -dependence of the cluster formation free energy $\Delta W_m = W_m - mW_1$ at various salt reservoir concentrations $\phi_{s,r}$. From the results shown in Figure 6b, we observe qualitatively similar behaviors to the salt-free case. Quantitatively, the magnitude of ΔW_m decreases with increasing $\phi_{s,r}$, reflecting the weakening of the driving force for coalescence with increasing salt. Following the same strategy as

in the salt-free case, we calculate the polycation concentration in the m -clusters, ϕ_m , and the total polycation concentration on the supernatant binodal, ϕ_p^I ; the results are shown in Figure 7a. Again, from the overlap between the black solid curve and the red dashed curve, we see that the contribution in the supernatant phase is dominated by the contribution from polyion pairs instead of larger clusters.

From Figure 5b, we see that the salt concentration inside the finite-sized cluster in the supernatant phase is slightly less than the salt concentration outside (the reservoir concentration) $\phi_{s,r}$. However, because the concentration of clusters in the supernatant phase is quite low, to a good approximation the salt concentration in the supernatant phase is the same as that in the reservoir, i.e., $\phi_s^I \approx \phi_{s,r}$. By collecting ϕ_p^{II} , ϕ_s^{II} , ϕ_p^I and ϕ_s^I , we can construct the binodal in the polymer–salt concentration (i.e., ϕ_p – ϕ_s) plane.

In Figure 7b, we show the binodal (the red solid curve). Similar to the salt-free case, we also sketch the missing part of the binodal close to the critical point, as shown by the red dotted curve, for a visual completeness of the phase diagram. For comparison, we include the binodal from the uniform mixing approximation (i.e., the black dotted curve), which is obtained by numerically solving the phase equilibrium equations (i.e., the equality of osmotic pressure, of the chemical potential of polyions, and small ions in the two coexisting phases) based on the Helmholtz free energy density given by eq 5. We note that the binodal reported in most of the previous theoretical studies is usually obtained under the uniform mixing approximation. Similar to the ϕ_p – l_B phase diagram for the salt-free case (i.e., Figure 3a), we find that the coacervate branch obtained from both methods is nearly the same, but the polyion concentration in the dilute branch in the cluster method is several orders of magnitude larger than that from the uniform mixing approximation. The magnitude of the polyion concentration in the supernatant branch of the binodal is in the experimentally observable range over a wide range of salt concentrations. This feature is qualitatively consistent with several experimental phase diagrams.^{33,49,60} Moreover, we expect the critical salt concentration in the method to be lower than predicted from the uniform mixing approximation (the black circle in Figure 7b). Finally, in the cluster treatment, the salt concentration in the supernatant phase is slightly higher than that in the coacervate phase over a wide range of salt concentrations (not shown); this behavior is similar to the salt partitioning obtained under the uniform mixing approximation. We note that it is possible to produce the opposite salt partition behavior when a specific polymer–salt attraction is included.^{62,75,137}

Using the same procedure described in the salt-free case, we also construct the pseudo-spinodal in the ϕ_p – ϕ_s plane to mark the effective limit of the metastable region; this is shown as the blue dashed-dotted curve in Figure 7b. For comparison, we also include the spinodal curve under the uniform mixing approximation (the green dotted curve), determined by the vanishing of the determinant of the Hessian matrix for the free energy density given by eq 5. We see that the pseudo-spinodal is in general larger than the uniform mixing spinodal. Based on the binodal and pseudo-spinodal in Figure 7b, we expect two types of phase separation kinetics depending on the initial polymer concentration ϕ_p . When ϕ_p is small (i.e., $5 \times 10^{-9} \lesssim \phi_p \lesssim 1.3 \times 10^{-4}$, bracketed by the starting points of the red solid and the blue dashed-dotted curve on the horizontal axis), phase separation proceeds via the nucleation and growth

kinetics under all salt concentrations $\phi_{s,r}$. On the other hand, when $\phi_p \gtrsim 1.3 \times 10^{-4}$, the phase separation mechanism is spinodal decomposition kinetics at low salt concentrations and changes over to nucleation and growth kinetics at increased salt concentrations.

IV. SUMMARY

In this work, using our recently developed mean-field theory for inhomogeneous polyelectrolyte solutions,⁸² we have examined the structure, interfacial tension, and formation free energy of symmetric finite-sized clusters in dilute solutions of oppositely charged polyions. These cluster properties are then used in combination with a simple thermodynamic theory for dilute solutions of clusters to examine their effects on the phase diagram. We find that the clusters in general consist of a uniform core and a diffusive interfacial layer. The polycation (or polyanion) concentration in the uniform core of a cluster having m polyion pairs, denoted by $\phi_{p,c}$, decreases with increasing m and follows $\phi_{p,c} = \phi_p^{II} (1 + C_p m^{-1/3})$ at large m , where ϕ_p^{II} is the polycation concentration in the coacervate phase and $C_p > 0$ is a constant that increases with decreasing l_B/σ or increasing $\phi_{s,r}$; this is a reflection of the Laplace pressure effect. Likewise, the interfacial tension of a finite-sized cluster, γ_m , decreases on increasing the cluster size and follows a similar scaling relation. Furthermore, for solutions with finite $\phi_{s,r}$ the small ions can be enriched in the interfacial regions of the droplet, and the strength of this enrichment decreases with increasing $\phi_{s,r}$.

Accounting for polyion pairs and clusters is shown to drastically affect the dilute branch of the phase diagram for symmetric polyelectrolyte complex coacervation. We construct the binodal and pseudo-spinodal in the ϕ_p – l_B plane for the salt-free solutions and in the ϕ_p – ϕ_s plane for solutions with salt. While the coacervate branch of the binodal using the current approach is almost the same as that obtained under the uniform mixing approximation, the polyion concentration in the supernatant branch from the method is several orders of magnitude higher than that from the uniform mixing approximation. These behaviors are qualitatively consistent with the field-theoretical results by Delaney and Fredrickson.⁴³ Based on the formation free energy, we further examine the nucleation free energy of a cluster from polyion pairs in a supersaturated solution as a function of the cluster size m and identify the nucleation free energy barrier ΔG^* and the critical nucleus size m^* . Setting this free energy barrier to order $10 k_B T$ allows us to obtain the pseudo-spinodal for the dilute solution. This pseudo-spinodal is shown to be also shifted significantly to higher polyion concentrations in comparison to the mean-field spinodal from the uniform mixing approximation, for both the salt-free solutions and the solutions with salt.

Making the volume approximation for the clusters, we also provide simple analytical expressions for the cluster formation free energy, the supernatant binodal, and the nucleation barrier. In particular, we propose a simple approximate expression for the polyion concentration in the coexisting supernatant phase, ϕ_p^I , in terms of the concentration of the coacervate phase, ϕ_p^{II} , and the interfacial tension for the planar interface γ_∞ , in the form of $k_B T \ln \phi_p^I \approx -(36\pi)^{1/3} (Nv/\phi_p^{II})^{2/3} \gamma_\infty$. This relation can be used as a useful order-of-magnitude estimate of ϕ_p^I using the more easily measurable properties ϕ_p^{II} and γ_∞ . Similarly, we are able to obtain simple expressions for the nucleation barrier ΔG^* and the size of the critical nucleus m^* in terms of ϕ_1/ϕ_p^I ,

chain length N , interfacial tension γ_{∞} , and polyion concentration in the coacervate phase ϕ_p^{II} .

By focusing on the supernatant phase, this study serves to provide a more complete picture on the global phase behavior of symmetric complex coacervation. On the other hand, the supernatant phase for asymmetric complex coacervation remains largely unexplored. Depending on the specific asymmetry conditions, the supernatant phase in equilibrium with an asymmetric coacervate may consist of finite-sized clusters as well as uncomplex polyions. In particular, the charge balance between polycation and polyanion can be violated inside the cluster, and correspondingly, the cluster usually carries a finite charge. When the amount of the finite charge is sufficiently high, the cluster may deviate from the spherical shape and adopt cylindrical or tadpole structures, as proposed by Oskolkov and Potemkin¹¹³ and by Zhang and Shklovskii,¹¹² respectively. A systematic investigation of the structure of such finite-sized clusters and their influence on the binodal and phase-transition kinetics for asymmetric complex coacervation represents an interesting future direction.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.2c00340>.

Expressions for the chemical potential of each species, the osmotic pressure, the binodal and spinodal under the uniform mixing approximation for the bulk solution; polyion concentration profiles and the formation free energy for a polyion pair without salt at various I_B/σ polyion and salt concentration profiles for a polyion pair at various $\phi_{s,r}$; and scalings of polyion and salt concentrations in the uniform core of various clusters (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Pengfei Zhang – State Key Laboratory for Modification of Chemical Fibers and Polymer Materials, Center for Advanced Low-Dimension Materials, College of Material Science and Engineering, Donghua University, Shanghai 201620, China; orcid.org/0000-0002-4226-1394; Email: pfzhangphy@dhu.edu.cn

Zhen-Gang Wang – Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California 91125, United States; orcid.org/0000-0002-3361-6114; Email: zgw@caltech.edu

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.macromol.2c00340>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors thank Prof. Isamu Kusaka for valuable discussion on the interfacial tension calculation for finite-sized droplets. P.Z. acknowledges the financial support provided by the National Natural Science Foundation of China (22073016 and 21803011) and the award of Shanghai Dongfang Scholar. Z.-G.W. acknowledges financial support from the Hong Kong Quantum AI Lab Ltd.

■ REFERENCES

- (1) de Jong, H. G. B.; Kruij, H. R. Coacervation (Partial Miscibility in Colloid Systems). *Chemistry* **1929**, *131*, 849–856.
- (2) Overbeek, J. T. G.; Voorn, M. J. Phase Separation in Polyelectrolyte Solutions. Theory of Complex Coacervation. *J. Cell. Comp. Physiol.* **1957**, *49*, 7–26.
- (3) Decher, G. Fuzzy Nanoassemblies: Toward Layered Polymeric Multicomposites. *Science* **1997**, *277*, 1232–1237.
- (4) Salehi, A.; Desai, P. S.; Li, J.; Steele, C. A.; Larson, R. G. Relationship between Polyelectrolyte Bulk Complexation and Kinetics of Their Layer-by-Layer Assembly. *Macromolecules* **2015**, *48*, 400–409.
- (5) Black, K. A.; Priftis, D.; Perry, S. L.; Yip, J.; Byun, W. Y.; Tirrell, M. Protein Encapsulation via Polypeptide Complex Coacervation. *ACS Macro Lett.* **2014**, *3*, 1088–1091.
- (6) Luo, F.; Sun, T. L.; Nakajima, T.; Kurokawa, T.; Ihsan, A. B.; Li, X.; Guo, H.; Gong, J. P. Free Reprocessability of Tough and Self-Healing Hydrogels Based on Polyion Complex. *ACS Macro Lett.* **2015**, *4*, 961–964.
- (7) Kelly, K. D.; Schlenoff, J. B. Spin-Coated Polyelectrolyte Coacervate Films. *ACS Appl. Mater. Interfaces* **2015**, *7*, 13980–13986.
- (8) Obermeyer, A. C.; Mills, C. E.; Dong, X.-H.; Flores, R. J.; Olsen, B. D. Complex Coacervation of Supercharged Proteins with Polyelectrolytes. *Soft Matter* **2016**, *12*, 3570–3581.
- (9) Meng, X.; Perry, S. L.; Schiffrin, J. D. Complex Coacervation: Chemically Stable Fibers Electrospun from Aqueous Polyelectrolyte Solutions. *ACS Macro Lett.* **2017**, *6*, 505–511.
- (10) Wan, A. C.; Cutiongco, M. F.; Tai, B. C.; Leong, M. F.; Lu, H. F.; Yim, E. K. Fibers by Interfacial Polyelectrolyte Complexation – Processes, Materials and Applications. *Mater. Today* **2016**, *19*, 437–450.
- (11) Srivastava, S.; Andreev, M.; Levi, A. E.; Goldfeld, D. J.; Mao, J.; Heller, W. T.; Prabhu, V. M.; de Pablo, J. J.; Tirrell, M. V. Gel Phase Formation in Dilute Triblock Copolyelectrolyte Complexes. *Nat. Commun.* **2017**, *8*, No. 14131.
- (12) Shi, R.; Sun, T. L.; Luo, F.; Nakajima, T.; Kurokawa, T.; Bin, Y. Z.; Rubinstein, M.; Gong, J. P. Elastic–Plastic Transformation of Polyelectrolyte Complex Hydrogels from Chitosan and Sodium Hyaluronate. *Macromolecules* **2018**, *51*, 8887–8898.
- (13) Zhao, M.; Zacharia, N. S. Protein Encapsulation via Polyelectrolyte Complex Coacervation: Protection against Protein Denaturation. *J. Chem. Phys.* **2018**, *149*, No. 163326.
- (14) Wang, J.; Sun, S.; Wu, B.; Hou, L.; Ding, P.; Guo, X.; Stuart, M. A. C.; Wang, J. Processable and Luminescent Supramolecular Hydrogels from Complex Coacervation of Polycations with Lanthanide Coordination Polyanions. *Macromolecules* **2019**, *52*, 8643–8650.
- (15) McTigue, W. C. B.; Perry, S. L. Protein Encapsulation Using Complex Coacervates: What Nature Has to Teach Us. *Small* **2020**, *16*, No. 1907671.
- (16) Brangwynne, C. P.; Eckmann, C. R.; Courson, D. S.; Rybarska, A.; Hoegge, C.; Gharakhani, J.; Julicher, F.; Hyman, A. A. Germline P Granules Are Liquid Droplets That Localize by Controlled Dissolution/Condensation. *Science* **2009**, *324*, 1729–1732.
- (17) Stewart, R. J.; Wang, C. S.; Shao, H. Complex Coacervates as a Foundation for Synthetic Underwater Adhesives. *Adv. Colloid Interface Sci.* **2011**, *167*, 85–93.
- (18) Aumiller, W. M.; Keating, C. D. Phosphorylation-Mediated RNA/Peptide Complex Coacervation as a Model for Intracellular Liquid Organelles. *Nat. Chem.* **2016**, *8*, 129–137.
- (19) Kim, S.; Huang, J.; Lee, Y.; Dutta, S.; Yoo, H. Y.; Jung, Y. M.; Jho, Y.; Zeng, H.; Hwang, D. S. Complexation and Coacervation of Like-Charged Polyelectrolytes Inspired by Mussels. *Proc. Natl. Acad. Sci. U.S.A.* **2016**, *113*, E847–E853.
- (20) Stewart, R. J.; Wang, C. S.; Song, I. T.; Jones, J. P. The Role of Coacervation and Phase Transitions in the Sandcastle Worm Adhesive System. *Adv. Colloid Interface Sci.* **2017**, *239*, 88–96.

- (21) Banani, S. F.; Lee, H. O.; Hyman, A. A.; Rosen, M. K. Biomolecular Condensates: Organizers of Cellular Biochemistry. *Nat. Rev. Mol. Cell Biol.* **2017**, *18*, 285–298.
- (22) Shin, Y.; Brangwynne, C. P. Liquid Phase Condensation in Cell Physiology and Disease. *Science* **2017**, *357*, No. eaaf4382.
- (23) Dompé, M.; Cedano-Serrano, F. J.; Heckert, O.; van den Heuvel, N.; van der Gucht, J.; Tran, Y.; Hourdet, D.; Creton, C.; Kamperman, M. Thermoresponsive Complex Coacervate-Based Underwater Adhesive. *Adv. Mater.* **2019**, *31*, No. 1808179.
- (24) Mountain, G. A.; Keating, C. D. Formation of Multiphase Complex Coacervates and Partitioning of Biomolecules within them. *Biomacromolecules* **2020**, *21*, 630–640.
- (25) Moreau, N. G.; Martin, N.; Gobbo, P.; Tang, T.-Y. D.; Mann, S. Spontaneous Membrane-less Multi-Compartmentalization via Aqueous Two-Phase Separation in Complex Coacervate Micro-Droplets. *Chem. Commun.* **2020**, *56*, 12717–12720.
- (26) Abbas, M.; Lipiński, W. P.; Wang, J.; Spruijt, E. Peptide-Based Coacervates as Biomimetic Protocells. *Chem. Soc. Rev.* **2021**, *50*, 3690–3705.
- (27) Fan, H.; Gong, J. P. Bioinspired Underwater Adhesives. *Adv. Mater.* **2021**, *33*, No. 2102983.
- (28) Castelnovo, M.; Joanny, J.-F. Complexation between Oppositely Charged Polyelectrolytes: Beyond RPA. *Eur. Phys. J. E* **2001**, *6*, 377–386.
- (29) Kudlay, A.; Ermoshkin, A. V.; de la Cruz, M. O. Complexation of Oppositely Charged Polyelectrolytes: Effect of Ion Pair Formation. *Macromolecules* **2004**, *37*, 9231–9241.
- (30) Kudlay, A.; de la Cruz, M. O. Precipitation of Oppositely Charged Polyelectrolytes in Salt Solutions. *J. Chem. Phys.* **2004**, *120*, 404–412.
- (31) Lee, J.; Popov, Y. O.; Fredrickson, G. H. Complex Coacervation: A Field Theoretic Simulation Study of Polyelectrolyte Complexation. *J. Chem. Phys.* **2008**, *128*, No. 224908.
- (32) Chollakup, R.; Smiththipong, W.; Eisenbach, C. D.; Tirrell, M. Phase Behavior and Coacervation of Aqueous Poly(acrylic acid)-Poly(allylamine) Solutions. *Macromolecules* **2010**, *43*, 2518–2528.
- (33) Spruijt, E.; Westphal, A. H.; Borst, J. W.; Stuart, M. A. C.; van der Gucht, J. Binodal Compositions of Polyelectrolyte Complexes. *Macromolecules* **2010**, *43*, 6476–6484.
- (34) Potemkin, I. I.; Palyulin, V. V. Complexation of Oppositely Charged Polyelectrolytes: Effect of Discrete Charge Distribution along the Chain. *Phys. Rev. E* **2010**, *81*, No. 041802.
- (35) Priftis, D.; Tirrell, M. Phase Behaviour and Complex Coacervation of Aqueous Polypeptide Solutions. *Soft Matter* **2012**, *8*, 9396–9405.
- (36) Spruijt, E.; Leermakers, F. A. M.; Fokkink, R.; Schweins, R.; van Well, A. A.; Stuart, M. A. C.; van der Gucht, J. Structure and Dynamics of Polyelectrolyte Complex Coacervates Studied by Scattering of Neutrons, X-rays, and Light. *Macromolecules* **2013**, *46*, 4596–4605.
- (37) Qin, J.; Priftis, D.; Farina, R.; Perry, S. L.; Leon, L.; Whitmer, J.; Hoffmann, K.; Tirrell, M.; de Pablo, J. J. Interfacial Tension of Polyelectrolyte Complex Coacervate Phases. *ACS Macro Lett.* **2014**, *3*, 565–568.
- (38) Wang, Q.; Schlenoff, J. B. The Polyelectrolyte Complex/Coacervate Continuum. *Macromolecules* **2014**, *47*, 3108–3116.
- (39) Perry, S. L.; Sing, C. E. PRISM-Based Theory of Complex Coacervation: Excluded Volume versus Chain Correlation. *Macromolecules* **2015**, *48*, 5040–5053.
- (40) Qin, J.; de Pablo, J. J. Criticality and Connectivity in Macromolecular Charge Complexation. *Macromolecules* **2016**, *49*, 8789–8800.
- (41) Salehi, A.; Larson, R. G. A Molecular Thermodynamic Model of Complexation in Mixtures of Oppositely Charged Polyelectrolytes with Explicit Account of Charge Association/Dissociation. *Macromolecules* **2016**, *49*, 9706–9719.
- (42) Rumyantsev, A. M.; Potemkin, I. I. Explicit Description of Complexation between Oppositely Charged Polyelectrolytes as an Advantage of the Random Phase Approximation over the Scaling Approach. *Phys. Chem. Chem. Phys.* **2017**, *19*, 27580–27592.
- (43) Delaney, K. T.; Fredrickson, G. H. Theory of Polyelectrolyte Complexation—Complex Coacervates are Self-Coacervates. *J. Chem. Phys.* **2017**, *146*, No. 224902.
- (44) Lytle, T. K.; Sing, C. E. Transfer Matrix Theory of Polymer Complex Coacervation. *Soft Matter* **2017**, *13*, 7001–7012.
- (45) Radhakrishna, M.; Basu, K.; Liu, Y.; Shamsi, R.; Perry, S. L.; Sing, C. E. Molecular Connectivity and Correlation Effects on Polymer Coacervation. *Macromolecules* **2017**, *50*, 3030–3037.
- (46) Chang, L.-W.; Lytle, T. K.; Radhakrishna, M.; Madinya, J. J.; Vélez, J.; Sing, C. E.; Perry, S. L. Sequence and Entropy-based Control of Complex Coacervates. *Nat. Commun.* **2017**, *8*, No. 1273.
- (47) Sadman, K.; Wang, Q.; Chen, Y.; Keshavarz, B.; Jiang, Z.; Shull, K. R. Influence of Hydrophobicity on Polyelectrolyte Complexation. *Macromolecules* **2017**, *50*, 9417–9426.
- (48) Liu, Y.; Winter, H. H.; Perry, S. L. Linear Viscoelasticity of Complex Coacervates. *Adv. Colloid Interface Sci.* **2017**, *239*, 46–60.
- (49) Li, L.; Srivastava, S.; Andreev, M.; Marciel, A. B.; de Pablo, J. J.; Tirrell, M. V. Phase Behavior and Salt Partitioning in Polyelectrolyte Complex Coacervates. *Macromolecules* **2018**, *51*, 2988–2995.
- (50) Adhikari, S.; Leaf, M. A.; Muthukumar, M. Polyelectrolyte Complex Coacervation by Electrostatic Dipolar Interactions. *J. Chem. Phys.* **2018**, *149*, No. 163308.
- (51) Rumyantsev, A. M.; Zhulina, E. B.; Borisov, O. V. Complex Coacervate of Weakly Charged Polyelectrolytes: Diagram of States. *Macromolecules* **2018**, *51*, 3788–3801.
- (52) Rubinstein, M.; Liao, Q.; Panyukov, S. Structure of Liquid Coacervates Formed by Oppositely Charged Polyelectrolytes. *Macromolecules* **2018**, *51*, 9572–9588.
- (53) Zhang, P.; Shen, K.; Alsaifi, N. M.; Wang, Z.-G. Salt Partitioning in Complex Coacervation of Symmetric Polyelectrolytes. *Macromolecules* **2018**, *51*, 5586–5593.
- (54) Zhang, P.; Alsaifi, N. M.; Wu, J.; Wang, Z.-G. Polyelectrolyte Complex Coacervation: Effects of Concentration Asymmetry. *J. Chem. Phys.* **2018**, *149*, No. 163303.
- (55) Shen, K.; Wang, Z.-G. Polyelectrolyte Chain Structure and Solution Phase Behavior. *Macromolecules* **2018**, *51*, 1706–1717.
- (56) Marciel, A. B.; Srivastava, S.; Tirrell, M. V. Structure and Rheology of Polyelectrolyte Complex Coacervates. *Soft Matter* **2018**, *14*, 2454–2464.
- (57) Hamad, F. G.; Chen, Q.; Colby, R. H. Linear Viscoelasticity and Swelling of Polyelectrolyte Complex Coacervates. *Macromolecules* **2018**, *51*, 5547–5555.
- (58) Andreev, M.; Prabhu, V. M.; Douglas, J. F.; Tirrell, M.; de Pablo, J. J. Complex Coacervation in Polyelectrolytes from a Coarse-Grained Model. *Macromolecules* **2018**, *51*, 6717–6723.
- (59) Fares, H. M.; Ghousoub, Y. E.; Delgado, J. D.; Fu, J.; Urban, V. S.; Schlenoff, J. B. Scattering Neutrons along the Polyelectrolyte Complex/Coacervate Continuum. *Macromolecules* **2018**, *51*, 4945–4955.
- (60) Lou, J.; Friedowitz, S.; Qin, J.; Xia, Y. Tunable Coacervation of Well-Defined Homologous Polyanions and Polycations by Local Polarity. *ACS Cent. Sci.* **2019**, *5*, 549–557.
- (61) Ali, S.; Bleuel, M.; Prabhu, V. M. Lower Critical Solution Temperature in Polyelectrolyte Complex Coacervates. *ACS Macro Lett.* **2019**, *8*, 289–293.
- (62) Schlenoff, J. B.; Yang, M.; Digby, Z. A.; Wang, Q. Ion Content of Polyelectrolyte Complex Coacervates and the Donnan Equilibrium. *Macromolecules* **2019**, *52*, 9149–9159.
- (63) Ali, S.; Prabhu, V. M. Characterization of the Ultralow Interfacial Tension in Liquid-Liquid Phase Separated Polyelectrolyte Complex Coacervates by the Deformed Drop Retraction Method. *Macromolecules* **2019**, *52*, 7495–7502.
- (64) Adhikari, S.; Prabhu, V. M.; Muthukumar, M. Lower Critical Solution Temperature Behavior in Polyelectrolyte Complex Coacervates. *Macromolecules* **2019**, *52*, 6998–7004.
- (65) Rumyantsev, A. M.; Jackson, N. E.; Yu, B.; Ting, J. M.; Chen, W.; Tirrell, M. V.; de Pablo, J. J. Controlling Complex Coacervation

- via Random Polyelectrolyte Sequences. *ACS Macro Lett.* **2019**, *8*, 1296–1302.
- (66) Lu, T.; Spruijt, E. Multiphase Complex Coacervate Droplets. *J. Am. Chem. Soc.* **2020**, *142*, 2905–2914.
- (67) Majee, A.; Bier, M.; Blosssey, R.; Podgornik, R. Charge Symmetry Broken Complex Coacervation. *Phys. Rev. Res.* **2020**, *2*, No. 043417.
- (68) Danielsen, S. P. O.; Panyukov, S.; Rubinstein, M. Ion Pairing and the Structure of Gel Coacervates. *Macromolecules* **2020**, *53*, 9420–9442.
- (69) Ye, Z.; Sun, S.; Wu, P. Distinct Cation–Anion Interactions in the UCST and LCST Behavior of Polyelectrolyte Complex Aqueous Solutions. *ACS Macro Lett.* **2020**, *9*, 974–979.
- (70) Li, L.; Romyantsev, A. M.; Srivastava, S.; Meng, S.; de Pablo, J. J.; Tirrell, M. V. Effect of Solvent Quality on the Phase Behavior of Polyelectrolyte Complexes. *Macromolecules* **2021**, *54*, 105–114.
- (71) Akkaoui, K.; Yang, M.; Digby, Z. A.; Schlenoff, J. B. Ultraviscosity in Entangled Polyelectrolyte Complexes and Coacervates. *Macromolecules* **2020**, *53*, 4234–4246.
- (72) Yu, B.; Rauscher, P. M.; Jackson, N. E.; Romyantsev, A. M.; de Pablo, J. J. Crossover from Rouse to Reptation Dynamics in Salt-Free Polyelectrolyte Complex Coacervates. *ACS Macro Lett.* **2020**, *9*, 1318–1324.
- (73) Liu, Y.; Chalarca, C. F. S.; Carmean, R. N.; Olson, R. A.; Madinya, J.; Sumerlin, B. S.; Sing, C. E.; Emrick, T.; Perry, S. L. Effect of Polymer Chemistry on the Linear Viscoelasticity of Complex Coacervates. *Macromolecules* **2020**, *53*, 7851–7864.
- (74) Audus, D. J.; Ali, S.; Romyantsev, A. M.; Ma, Y.; de Pablo, J. J.; Prabhu, V. M. Molecular Mass Dependence of Interfacial Tension in Complex Coacervation. *Phys. Rev. Lett.* **2021**, *126*, No. 237801.
- (75) Neitzel, A. E.; Fang, Y. N.; Yu, B.; Romyantsev, A. M.; de Pablo, J. J.; Tirrell, M. V. Polyelectrolyte Complex Coacervation across a Broad Range of Charge Densities. *Macromolecules* **2021**, *54*, 6878–6890.
- (76) Friedowitz, S.; Lou, J.; Barker, K. P.; Will, K.; Xia, Y.; Qin, J. Looping-in Complexation and Ion Partitioning in Nonstoichiometric Polyelectrolyte Mixtures. *Sci. Adv.* **2021**, *7*, No. eabg8654.
- (77) Chen, X.; Chen, E.-Q.; Shi, A.-C.; Yang, S. Multiphase Coacervates Driven by Electrostatic Correlations. *ACS Macro Lett.* **2021**, *10*, 1041–1047.
- (78) Zheng, B.; Avni, Y.; Andelman, D.; Podgornik, R. Phase Separation of Polyelectrolytes: The Effect of Charge Regulation. *J. Phys. Chem. B* **2021**, *125*, 7863–7870.
- (79) Huang, J.; Laaser, J. E. Charge Density and Hydrophobicity-Dominated Regimes in the Phase Behavior of Complex Coacervates. *ACS Macro Lett.* **2021**, *10*, 1029–1034.
- (80) Aponte-Rivera, C.; Rubinstein, M. Dynamic Coupling in Unentangled Liquid Coacervates Formed by Oppositely Charged Polyelectrolytes. *Macromolecules* **2021**, *54*, 1783–1800.
- (81) Bobbili, S. V.; Milner, S. T. A Simple Simulation Model for Complex Coacervates. *Soft Matter* **2021**, *17*, 9181–9188.
- (82) Zhang, P.; Wang, Z.-G. Interfacial Structure and Tension of Polyelectrolyte Complex Coacervates. *Macromolecules* **2021**, *54*, 10994–11007.
- (83) Ylitalo, A. S.; Balzer, C.; Zhang, P.; Wang, Z.-G. Electrostatic Correlations and Temperature-Dependent Dielectric Constant Can Model LCST in Polyelectrolyte Complex Coacervation. *Macromolecules* **2021**, *54*, 11326–11337.
- (84) Ma, Y.; Ali, S.; Prabhu, V. M. Enhanced Concentration Fluctuations in Model Polyelectrolyte Coacervate Mixtures along a Salt Isopleth Phase Diagram. *Macromolecules* **2021**, *54*, 11338–11350.
- (85) van der Gucht, J.; Spruijt, E.; Lemmers, M.; Stuart, M. A. C. Polyelectrolyte Complexes: Bulk Phases and Colloidal Systems. *J. Colloid Interface Sci.* **2011**, *361*, 407–422.
- (86) Veis, A. A Review of the Early Development of the Thermodynamics of the Complex Coacervation Phase Separation. *Adv. Colloid Interface Sci.* **2011**, *167*, 2–11.
- (87) Srivastava, S.; Tirrell, M. V. Polyelectrolyte Complexation. In *Advances in Chemical Physics*, John Wiley & Sons, Inc., 2016; pp 499–544.
- (88) Sing, C. E. Development of the Modern Theory of Polymeric Complex Coacervation. *Adv. Colloid Interface Sci.* **2017**, *239*, 2–16.
- (89) Muthukumar, M. 50th Anniversary Perspective: A Perspective on Polyelectrolyte Solutions. *Macromolecules* **2017**, *50*, 9528–9560.
- (90) Sing, C. E.; Perry, S. L. Recent Progress in the Science of Complex Coacervation. *Soft Matter* **2020**, *16*, 2885–2914.
- (91) Lalwani, S. M.; Eneh, C. I.; Lutkenhaus, J. L. Emerging trends in the dynamics of polyelectrolyte complexes. *Phys. Chem. Chem. Phys.* **2020**, *22*, 24157–24177.
- (92) Romyantsev, A. M.; Jackson, N. E.; de Pablo, J. J. Polyelectrolyte Complex Coacervates: Recent Developments and New Frontiers. *Annu. Rev. Condens. Matter Phys.* **2021**, *12*, 155–176.
- (93) Prabhu, V. M. Interfacial Tension in Polyelectrolyte Systems Exhibiting Associative Liquid-Liquid Phase Separation. *Curr. Opin. Colloid Interface Sci.* **2021**, *53*, No. 101422.
- (94) Larson, R. G.; Liu, Y.; Li, H. Linear Viscoelasticity and Time-Temperature-Salt and other Superpositions in Polyelectrolyte Coacervates. *J. Rheol.* **2021**, *65*, 77–102.
- (95) Cahn, J. W.; Hilliard, J. E. Free Energy of a Nonuniform System. III. Nucleation in a Two-Component Incompressible Fluid. *J. Chem. Phys.* **1959**, *31*, 688–699.
- (96) Cahn, J. W. Phase Separation by Spinodal Decomposition in Isotropic Systems. *J. Chem. Phys.* **1965**, *42*, 93–99.
- (97) Shimobayashi, S. F.; Ronceray, P.; Sanders, D. W.; Haataja, M. P.; Brangwynne, C. P. Nucleation Landscape of Biomolecular Condensates. *Nature* **2021**, *599*, 503–506.
- (98) Veis, A.; Aranyi, C. Phase Separation In Polyelectrolyte Systems. I. Complex Coacervates of Gelatin. *J. Phys. Chem. A* **1960**, *64*, 1203–1210.
- (99) des Cloizeaux, J.; Jannink, G. *Polymers in Solution: Their Modelling and Structure*, Clarendon Press, 1991.
- (100) Grosberg, A. Y.; Kuznetsov, D. V. Quantitative theory of the globule-to-coil transition. 1. Link density distribution in a globule and its radius of gyration. *Macromolecules* **1992**, *25*, 1970–1979.
- (101) Grosberg, A. Y.; Kuznetsov, D. V. Quantitative theory of the globule-to-coil transition. 2. Density-density correlation in a globule and the hydrodynamic radius of a macromolecule. *Macromolecules* **1992**, *25*, 1980–1990.
- (102) Grosberg, A. Y.; Kuznetsov, D. V. Quantitative theory of the globule-to-coil transition. 3. Globule-globule interaction and polymer solution binodal and spinodal curves in the globular range. *Macromolecules* **1992**, *25*, 1991–1995.
- (103) Grosberg, A. Y.; Kuznetsov, D. V. Quantitative theory of the globule-to-coil transition. 4. Comparison of theoretical results with experimental data. *Macromolecules* **1992**, *25*, 1996–2003.
- (104) Wang, R.; Wang, Z.-G. Theory of Polymers in Poor Solvent: Phase Equilibrium and Nucleation Behavior. *Macromolecules* **2012**, *45*, 6266–6271.
- (105) Wang, R.; Wang, Z.-G. Theory of Polymer Chains in Poor Solvent: Single-Chain Structure, Solution Thermodynamics, and Θ Point. *Macromolecules* **2014**, *47*, 4094–4102.
- (106) Wang, Z.-G. 50th Anniversary Perspective: Polymer Conformation—A Pedagogical Review. *Macromolecules* **2017**, *50*, 9073–9114.
- (107) Lifshitz, I. M. Some Problems of the Statistical Theory of Biopolymers. *Sov. Phys. JETP* **1969**, *28*, 1280.
- (108) Borue, V. Y.; Erukhimovich, I. Y. A Statistical Theory of Globular Polyelectrolyte Complexes. *Macromolecules* **1990**, *23*, 3625–3632.
- (109) Tainaka, K.-I. Study of Complex Coacervation in Low Concentration by Virial Expansion Method. I. Salt Free Systems. *J. Phys. Soc. Jpn.* **1979**, *46*, 1899–1906.
- (110) Tainaka, K.-I. Effect of Counterions on Complex Coacervation. *Biopolymers* **1980**, *19*, 1289–1298.

- (111) Tainaka, K.-I. Study of Complex Coacervation in Low Concentration by Virial Expansion Method. II. Salt-Containing Systems. *J. Phys. Soc. Jpn.* **1983**, *52*, 2603–2609.
- (112) Zhang, R.; Shklovskii, B. Phase Diagram of Solution of Oppositely Charged Polyelectrolytes. *Phys. A* **2005**, *352*, 216–238.
- (113) Oskolkov, N. N.; Potemkin, I. I. Complexation in Asymmetric Solutions of Oppositely Charged Polyelectrolytes: Phase Diagram. *Macromolecules* **2007**, *40*, 8423–8429.
- (114) Hayashi, Y.; Ullner, M.; Linse, P. A Monte Carlo study of solutions of oppositely charged polyelectrolytes. *J. Chem. Phys.* **2002**, *116*, 6836–6845.
- (115) Ou, Z.; Muthukumar, M. Entropy and enthalpy of polyelectrolyte complexation: Langevin dynamics simulations. *J. Chem. Phys.* **2006**, *124*, No. 154902.
- (116) Zhao, M.; Zhou, J.; Su, C.; Niu, L.; Liang, D.; Li, B. Complexation behavior of oppositely charged polyelectrolytes: Effect of charge distribution. *J. Chem. Phys.* **2015**, *142*, No. 204902.
- (117) Zhou, J.; Barz, M.; Schmid, F. Complex formation between polyelectrolytes and oppositely charged oligoelectrolytes. *J. Chem. Phys.* **2016**, *144*, No. 164902.
- (118) Xu, X.; Kanduć, M.; Wu, J.; Dzubiella, J. Potential of mean force and transient states in polyelectrolyte pair complexation. *J. Chem. Phys.* **2016**, *145*, No. 034901.
- (119) Rathee, V. S.; Zervoudakis, A. J.; Sidky, H.; Sikora, B. J.; Whitmer, J. K. Weak polyelectrolyte complexation driven by associative charging. *J. Chem. Phys.* **2018**, *148*, No. 114901.
- (120) Singh, A. N.; Yethiraj, A. Driving Force for the Complexation of Charged Polypeptides. *J. Phys. Chem. B* **2020**, *124*, 1285–1292.
- (121) Hill, T. L. *An Introduction to Statistical Thermodynamics*, Dover Publications, 1986.
- (122) Michaels, A. S. Polyelectrolyte Complexes. *Ind. Eng. Chem.* **1965**, *57*, 32–40.
- (123) In principle, one also needs to include the solvent reservoir term. However, because of the incompressibility, the solvent chemical potential can be absorbed into the chemical potential of the salt ions and the chemical potential of the polyions.
- (124) Mcquarrie, D. A. *Statistical Mechanics*, 1st ed.; University Science Books, 2000.
- (125) Rumyantsev, A. M.; Kramarenko, E. Y.; Borisov, O. V. Microphase Separation in Complex Coacervate Due to Incompatibility between Polyanion and Polycation. *Macromolecules* **2018**, *51*, 6587–6601.
- (126) Lifshitz, I. M.; Grosberg, A. Y.; Khokhlov, A. R. Some problems of the statistical physics of polymer chains with volume interaction. *Rev. Mod. Phys.* **1978**, *50*, 683–713.
- (127) Kusaka, I. *Statistical Mechanics for Engineers*, Springer International Publishing, 2015.
- (128) Tolman, R. C. The Effect of Droplet Size on Surface Tension. *J. Chem. Phys.* **1949**, *17*, 333–337.
- (129) This approximation ignores the small contribution from the low concentration of clusters to the osmotic pressure in the supernatant phase, which can be easily included, resulting in a self-consistent set of equations.
- (130) de Gennes, P.-G.; Brochard-Wyart, F.; Quéré, D. *Capillarity and Wetting Phenomena*, Springer: New York, 2004.
- (131) Blokhuis, E. M.; Kuipers, J. Thermodynamic expressions for the Tolman length. *J. Chem. Phys.* **2006**, *124*, No. 074701.
- (132) Kardar, M. *Statistical Physics of Fields*, Cambridge University Press, 2007.
- (133) Binder, K. Collective Diffusion, Nucleation, and Spinodal Decomposition in Polymer Mixtures. *J. Chem. Phys.* **1983**, *79*, 6387–6409.
- (134) Binder, K. Nucleation Barriers, Spinodals, and the Ginzburg Criterion. *Phys. Rev. A* **1984**, *29*, 341–349.
- (135) Wood, S. M.; Wang, Z.-G. Nucleation in binary polymer blends: A self-consistent field study. *J. Chem. Phys.* **2002**, *116*, 2289–2300.
- (136) Wang, Z.-G. Concentration Fluctuation in Binary Polymer Blends: χ Parameter, Spinodal and Ginzburg Criterion. *J. Chem. Phys.* **2002**, *117*, 481–500.
- (137) Ghasemi, M.; Friedowitz, S.; Larson, R. G. Analysis of Partitioning of Salt through Doping of Polyelectrolyte Complex Coacervates. *Macromolecules* **2020**, *53*, 6928–6945.

Recommended by ACS

Unifying Weak and Strong Charge Correlations within the Random Phase Approximation: Polyampholytes of Various Sequences

Artem M. Rumyantsev, Juan J. de Pablo, *et al.*

JULY 06, 2022
MACROMOLECULES

READ 

Phase Behaviors of a Binary Blend of Oppositely Charged Polyelectrolytes: A Weak Segregation Approach

Mingge Zhao, Junhan Cho, *et al.*

AUGUST 16, 2022
MACROMOLECULES

READ 

Complexation between Oppositely Charged Polyelectrolytes in Dilute Solution: Effects of Charge Asymmetry

Shensheng Chen, Zhen-Gang Wang, *et al.*

MAY 13, 2022
MACROMOLECULES

READ 

Effects of Ion Valency on Polyelectrolyte Brushes: A Unified Theory

Minglun Li, Jing Yu, *et al.*

NOVEMBER 16, 2022
MACROMOLECULES

READ 

Get More Suggestions >